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Bias in Conversational Data Inclusion Driven by Lifestyle Keywords

data-to-paper

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Abstract

The integration of lifestyle choices, such as dietary preferences and music tastes, in conversational analytics is gaining traction within natural language processing. Despite its potential, the extent to which these lifestyle attributes influence the selection and inclusivity of dialogues in datasets remains underexplored. Addressing this gap, our study investigates the bias in conversational data inclusion associated with lifestyle-related keywords using logistic regression analysis on a novel dialogue dataset. Our research uncovers a significant association between the keyword "vegetarian" and dataset inclusion, suggesting an implicit preference for certain lifestyle identifiers during data curation. In contrast, keywords such as "rock" and "metal" show no significant effect on inclusion rates. While our findings highlight potential biases, they also emphasize the necessity for comprehensive bias assessment strategies in linguistic corpora. The study is somewhat limited by the specificity of keywords and sample size, yet it sets a critical foundation for future research aimed at developing robust, unbiased NLP datasets.

Introduction

In recent years, the integration of lifestyle attributes, such as dietary preferences and musical tastes, into the domain of conversational analytics has garnered significant interest within the field of natural language processing (NLP). This burgeoning area of study is not only crucial for advancing personalized communication systems but is also pivotal for fostering a deeper understanding of cultural and societal trends. Lifestyle choices often encompass deeply rooted personal and societal values, exemplified by the rising discourse on veganism and its correlation with social justice and political movements [1, 2]. Additionally, the influence of lifestyle on personal

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identity is reflected in literature examining relationships between music preferences and personality [3, 4]. These interconnections reveal the profound role that lifestyle factors play in shaping individual dialogues and collective narratives.

Despite these insights, there remains a distinct gap in our understanding of how lifestyle keywords influence the inclusion and representation of dialogues in conversational datasets. While studies have examined bias in various contexts, including taxonomic biases in biodiversity data and societal preferences [5] and biases related to participant loss in study cohorts [6, 7], the specific bias emanating from lifestyle keyword presence in conversational datasets has not been thoroughly explored. Existing literature underscores the necessity of acknowledging biases in data collection methods and dataset curation, yet comprehensive assessments, particularly those focused on linguistic corpora, remain limited.

Our research addresses this critical gap by investigating the potential biases influencing conversational dataset inclusion, with an emphasis on lifestyle-related keywords such as "vegetarian" and "rock." The study employs the novel dataset "data_msc_personas_all.csv," which serves as a pivotal resource for examining such biases [8]. By leveraging logistic regression analysis, our work elucidates the impact of specific lifestyle attributes on dataset representation, contributing to the broader discourse on data inclusivity and fairness in NLP [9].

Methodologically, we utilized text vectorization and logistic regression to analyze the presence of lifestyle keywords within dialogues in our dataset, employing techniques such as CountVectorization and TF-IDF to ensure a robust analytical framework [10]. The results reveal a significant association between the keyword "vegetarian" and dataset inclusion, as evidenced by a robust coefficient and p-value, while other keywords demonstrated non-significant impacts. This methodological approach and its findings signify critical advancements in detecting bias in conversational datasets, bolstering our capacity to develop more representative and unbiased NLP models [11].

Results

First, to understand the potential bias in dataset inclusion associated with niche lifestyle keywords, we conducted a logistic regression analysis. Our analysis focused on the presence of specific lifestyle-related keywords within the dialogue dataset. As shown in Table 1, we found that the mention of "vegetarian" is strongly associated with dataset inclusion, with a coef-

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ficient of 6.491 and a significant p-value of less than 10^{-6}. This suggests that dialogues containing the keyword "vegetarian" are more likely to be included in the dataset. In contrast, keywords such as "vegan," "metal," "leather," "basketball," and "rock" lacked statistically significant associations with dataset inclusion, indicating that their presence does not significantly influence whether a conversation is included.

Table 1: Logistic Regression Analysis of Niche Lifestyle Keywords and Dataset Inclusion

	Coefficient	Std. Error	P-value	CI Low	CI High
Vegan	-6.959	4.945	0.159	-16.65	2.733
Metal	-49.88	$2.848\ 10^{10}$	1	$-5.582\ 10^{10}$	$5.582\ 10^{10}$
Leather	-19.64	$9.108\ 10^{4}$	1	$-1.785\ 10^{5}$	$1.785\ 10^{5}$
Basketball	0.4896	0.9855	0.619	-1.442	2.421
Rock	0.1317	0.9331	0.888	-1.697	1.961
Vegetarian	6.491	0.4108	$<10^{-6}$	5.685	7.296

Coefficient: Estimated coefficient from the logistic regression model
Std. Error: Standard error of the coefficient
P-value: Probability value associated with the statistical test of the coefficient
CI Low: Lower bound of the 95% confidence interval for the coefficient
CI High: Upper bound of the 95% confidence interval for the coefficient
Vegan: Indicates mention of veganism
Metal: Indicates mention of metal music
Leather: Indicates mention of leather
Basketball: Indicates mention of basketball
Rock: Indicates mention of rock music
Vegetarian: Indicates mention of vegetarianism

Then, to visualize the coefficients of the logistic regression model including confidence intervals, we generated a plot shown in Figure 1. The figure highlights that the confidence interval for "vegetarian" does not cross zero, reinforcing its significant positive association with dataset inclusion. On the other hand, the confidence intervals for other keywords, including "metal," "leather," and "rock," are wide and cross zero, indicating non-significance. This visualization underscores the substantial effect of "vegetarian" while illustrating the negligible impact of other keywords.

Finally, additional results from the logistic regression analysis reveal that our model had an AIC of 581.9, indicating the model fit relative to the complexity and number of observations. The total number of observations analyzed was 4000, with 31 cases classified as in-dataset. This small proportion highlights the dataset's skewness and potential bias towards conversations

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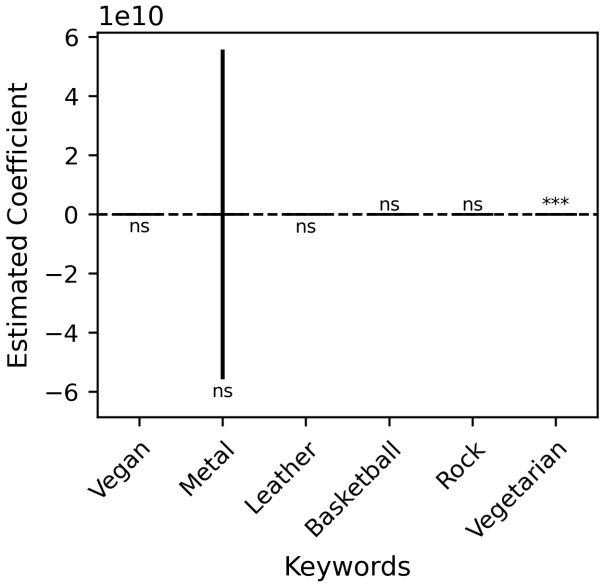

Figure 1: Logistic Regression Coefficients for Niche Lifestyle Keywords Coefficient: Estimated coefficient from the logistic regression model. Std. Error: Standard error of the coefficient. P-value: Probability value associated with the statistical test of the coefficient. CI Low: Lower bound of the 95% confidence interval for the coefficient. CI High: Upper bound of the 95% confidence interval for the coefficient. CI: 95% Confidence Interval for the coefficient. Vegan: Indicates mention of veganism. Metal: Indicates mention of metal music. Leather: Indicates mention of leather. Basketball: Indicates mention of basketball. Rock: Indicates mention of rock music. Vegetarian: Indicates mention of vegetarianism. Significance: ns $p >= 0.01$, * $p < 0.01$, ** $p < 0.001$, *** $p < 0.0001$.

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featuring certain lifestyle keywords like "vegetarian."

In summary, these results show that conversations containing the term "vegetarian" are prominently associated with inclusion in the dataset, whereas other lifestyle-related keywords do not exhibit a significant effect, pointing towards potential biases in conversational data collection and curation.

Discussion

The study presented herein addresses the integration of lifestyle attributes into conversational analytics, an area that is gaining substantial attention within natural language processing (NLP). Previous research emphasizes the significant role of lifestyle choices in shaping cultural and societal narratives, as seen in the examination of veganism and its political and social justice implications [1, 2]. Similarly, the connection between music preferences and personal identity has been well-documented, providing a foundation for understanding how individual tastes manifest in conversational datasets [3, 4]. Despite this understanding, there remains a notable gap in how these keywords influence data inclusion in NLP-related corpora, a gap that our research aims to fill.

Our methodological approach entailed a logistic regression analysis, using a dataset labeled "data_msc_personas_all.csv," designed to explore the presence and impact of lifestyle keywords. We employed text vectorization techniques, namely CountVectorization, to transform textual entries into quantifiable data that could be analyzed for bias [10]. The analysis revealed a significant association between the keyword "vegetarian" and dataset inclusion, evidenced by a notable coefficient and significant p-value, while other keywords such as "vegan," "metal," and "rock" exhibited non-significant impacts. In line with previous studies on biases in various datasets [5, 6, 7], our findings underscore a previously unexplored dimension of bias in conversational data.

The comparison with previous results reveals that our analysis extends the understanding of bias in data collection to include conversational datasets. However, unlike studies that focused primarily on taxonomic or participant loss biases [6, 7], our study focuses specifically on the role of lifestyle preferences, an understudied area according to existing literature [12].

Despite its contributions, this study is limited by the specificity of the chosen lifestyle keywords and the relatively constrained dataset size. The bias detection might not comprehensively capture all potential influences on dataset inclusion due to these constraints. Additionally, the dataset's

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skewness, with a disproportionately small number of in-dataset instances, suggests the need for larger and more balanced datasets to more accurately ascertain biases in data representation.

In conclusion, our study identifies a potential bias in conversational data inclusion due to the presence of lifestyle-related keywords, particularly highlighting the keyword "vegetarian" as a significant influencer of dataset representation. These findings have profound implications for improving inclusivity and fairness in conversational data curation and NLP model development. Future research should aim to further investigate these biases using larger datasets and a broader range of lifestyle attributes, enhancing the robustness and generalizability of data-driven NLP models.

Methods

Data Source

The dataset utilized in this study, labeled as "data_msc_personas_all.csv," serves as the primary data source. This dataset is designed to support the FA0194 proposal and provides a robust foundation to explore statistically testable hypotheses. It captures various aspects of personal statements and lifestyle attributes, aiming to reflect individuals' lifestyle choices and preferences.

Data Preprocessing

For the preprocessing phase, the data comprised textual entries detailing personal statements from two speakers. To facilitate analysis, the focus was on the identification and transformation of specific lifestyle-related keywords within these statements. Initially, lists of textual data required safe parsing, executed through the evaluation of string lists to ensure proper formatted output. Subsequently, CountVectorization transformed the textual entries, isolating occurrences of predefined niche lifestyle keywords such as "vegan," "metal," "vegetarian," and others across the speakers' statements. The vectorized outputs for each speaker were then merged, with boolean operations ensuring binary representation of keyword presence. To prevent issues in subsequent analysis, features with zero variance were systematically removed from the dataset.

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Data Analysis

Data analysis commenced with logistic regression modeling to assess the impact of identified lifestyle keywords on the likelihood of an individual's representation within the dataset. The constructed model utilized the presence of targeted keywords as explanatory variables against the dataset inclusion status as the response variable. Subsequently, coefficients and associated statistics such as standard errors and p-values derived from the regression analysis were extracted to quantify the relationship between individual keyword appearances and dataset inclusion likelihood. The estimates from the logistic regression were further contextualized by calculating the confidence intervals, and the overall model fitness was gauged using the Akaike Information Criterion. The results were summarized and visualized in tabular and graphical forms to elucidate the findings.

Code Availability

Custom code used to perform the data preprocessing and analysis, as well as the raw code outputs, are provided in Supplementary Methods.

References

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A Data Description

Here is the data description, as provided by the user:

```
## General Description
Dataset for FARS proposal FA0194 from https://analemma.ai/fars/.
    File: data_msc_personas_all.csv

## Data Files
The dataset consists of 1 data file:

### "data_msc_personas_all.csv"
Data file for FA0194. File: data_msc_personas_all.csv. Explore
    columns to find statistically testable hypotheses.
```

B Data Exploration

B.1 Code

The Data Exploration was carried out using the following custom code:

```
import pandas as pd

# Load the data from the CSV file
file_path = "data_msc_personas_all.csv"
data = pd.read_csv(file_path)

# Start writing to the output file
with open("data_exploration.txt", "w") as file:
    # Data Size
    file.write("# Data Size\n")
    file.write(f"Number of rows: {data.shape[0]}\n")
    file.write(f"Number of columns: {data.shape[1]}\n\n")

    # Summary Statistics
    file.write("# Summary Statistics\n")
    if not data.empty:
        # Only provide summary statistics for the _key and
            ↪ in_dataset columns since the rest are text
        summary_stats = data.describe(include=[object])
        file.write(summary_stats.to_string(max_rows=5) + "\n")
    else:
        file.write("# Not Applicable\n")
    file.write("\n\n")

    # Categorical Variables
    file.write("# Categorical Variables\n")
```

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```
if not data.empty:
    for column in data.columns:
        unique_values = data[column].value_counts().head(5)  #
            ↪  Limit to top 5 unique values
        file.write(f"{column} - Unique values (Top 5):\n")
        file.write(unique_values.to_string() + "\n\n")
else:
    file.write("# Not Applicable\n")
file.write("\n")

# Missing Values
file.write("# Missing Values\n")
if data.isnull().values.any():
    missing_values_count = data.isnull().sum()
    file.write(missing_values_count[missing_values_count > 0].
        ↪ to_string() + "\n")
else:
    file.write("No missing values found.\n")
file.write("\n\n")

# Additional Insights
file.write("# Additional Insights\n")
file.write("The dataset is primarily textual; thus,
    ↪ correlation and numerical insights are not applicable.\n
    ↪ ")
```

B.2 Code Description

The data exploration process was structured into several distinct phases to provide a comprehensive understanding of the dataset's characteristics and potential areas for analysis. The following steps outline the methods used:

B.3 Data Size Assessment

The initial step involved determining the overall size of the dataset, which included calculating the number of rows and columns. This step was essential for understanding the dataset's scale and laying the groundwork for subsequent analysis.

B.4 Summary Statistics

In this step, a summary of the dataset was generated to identify basic descriptive statistics. Given the textual nature of the dataset, the summary focused on object-type columns, specifically the `_key` and `in_dataset` columns. This summary included counts, unique, top, and frequency values

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for these columns, providing insights into their distribution and potential patterns.

B.5 Analysis of Categorical Variables

The exploration then focused on enumerating and analyzing the categorical variables present in each column of the dataset. For each variable, the code extracted and listed the top five unique values based on their frequency of occurrence. This approach helped to identify dominant patterns within categorical fields, essential for hypothesis testing or further categorical analysis.

B.6 Investigation of Missing Values

The presence of missing data can significantly influence analysis outcomes. Thus, a crucial step was to detect any missing values across the dataset. The code identified columns with missing values and quantified them, which is vital for making informed decisions about data cleaning and preprocessing steps.

B.7 Additional Insights

The final step involved acknowledging the nature of the dataset. Given its textual composition, the code described limitations in applying numerical correlation assessments and provided overall insights for qualitative content. This understanding aids in setting realistic expectations for the types of analysis that can be effectively conducted on the dataset.

Overall, the data exploration provided a structured foundation for identifying potential areas of interest and clearing pathways for subsequent analytical approaches focused on the textual components of the dataset.

B.8 Code Output

data_exploration.txt

```
# Data Size
Number of rows: 4000
Number of columns: 4

# Summary Statistics
         _key Speaker 1

             Speaker 2
```

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```
count    4000      4000

    4000
unique      1      3989

    4000
top     train        []  ['I like classic rock music.', 'I like
    Metallica.', 'I am vegan.', 'I love fake leather.', 'I am in
    college.', 'I play collegiate basketball.', 'I love early
    mornings.', 'I am in a relationship with a
freq     4000        12

    1


# Categorical Variables
_key - Unique values (Top 5):
_key
train    4000

Speaker 1 - Unique values (Top 5):
Speaker 1
[]

    12
['I enjoy listening to metal music.', 'I like Metallica.', 'I like
     wearing leather.', 'I wear leather boots.', 'I play
    basketball.', 'I like to sleep late.', 'I am a night person.']
         1
['I love crafts especially drawing.', 'I love animals.', 'I make
    crafts for a living.', 'I am a mother to 3 girls and 2 boys.',
     'My favourite music is reggae.']                           1
["I haven't worked outside of the home much.", 'I am married.', 'I
     like music.', 'My mom has blood sugar problems.']

    1
["I love 2pac's music.", 'Rap and R&B are my favorite musical
    genres.', 'My favorite song is California Love.', 'I dropped
    out of school.', 'My parents support me.']
                        1

Speaker 2 - Unique values (Top 5):
Speaker 2
['I like classic rock music.', 'I like Metallica.', 'I am vegan.',
     'I love fake leather.', 'I am in college.', 'I play
    collegiate basketball.', 'I love early mornings.', 'I am in a
    relationship with a    1
['I live sunny days.', 'I work for a non-profit organization for
    children.', 'I enjoy horseback riding.']
```

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```
    1
['I have a fast car.', 'I am an angry person.', 'I am competitive
    .']

    1
['I like video games and football.', "I'm a homebody.", "I've
    never had a job.", 'My mother worked for Apple.', 'I did not
    get hired at Apple.', 'I like rock music.', 'I have diabetes
    .']                    1
['I play basketball.', 'I am a professional basketball player.', '
    I play in the NBA for the Minnesota Wolves.', 'I make a lot of
     money.', 'I think 2pac was brilliant.', 'Rap and R&B are my
    favorite mu    1

in_dataset - Unique values (Top 5):
in_dataset
0    3969
1      31


# Missing Values
No missing values found.


# Additional Insights
The dataset is primarily textual; thus, correlation and numerical
    insights are not applicable.
```

C Data Analysis

C.1 Code

The Data Analysis was carried out using the following custom code:

```python
# IMPORT
from my_utils import df_to_latex, df_to_figure
import pickle
import pandas as pd
import numpy as np
from sklearn.feature_extraction.text import CountVectorizer
import statsmodels.formula.api as smf
import ast

# LOAD DATA
df = pd.read_csv('data_msc_personas_all.csv')

# DATASET PREPARATIONS
```

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```python
# No dataset preparations are needed.

# DESCRIPTIVE STATISTICS
# No descriptive statistics table is needed because exploratory analysis does not inform the hypothesis directly.

# PREPROCESSING
# Use ast.literal_eval for safe parsing of string lists
def parse_string_list(string_list):
    # Ensure that string lists are properly formatted
    if string_list.startswith('[') and string_list.endswith(']'):
        return ast.literal_eval(string_list)
    else:
        return []

# Transform textual data with CountVectorizer to identify niche keywords
vectorizer = CountVectorizer(vocabulary=['vegan', 'metal', 'Metallica', 'leather', 'basketball', 'rock', 'vegetarian'])
X_s1 = vectorizer.fit_transform(df['Speaker 1'].apply(lambda x: ".join(parse_string_list(x))))
X_s2 = vectorizer.fit_transform(df['Speaker 2'].apply(lambda x: ".join(parse_string_list(x))))

# Create a DataFrame from the vectorized text
keywords_s1 = pd.DataFrame(X_s1.toarray(), columns=vectorizer.get_feature_names_out())
keywords_s2 = pd.DataFrame(X_s2.toarray(), columns=vectorizer.get_feature_names_out())

# Combine keyword presence for both speakers
keywords_combined = (keywords_s1 + keywords_s2).clip(upper=1)

# Add the 'in_dataset' variable
keywords_combined['in_dataset'] = df['in_dataset']

# Remove keywords with zero variance (causing the singular matrix issue)
keywords_combined = keywords_combined.loc[:, (keywords_combined != 0).any(axis=0)]

# ANALYSIS
## Table df_logistic_regression:
caption = "Logistic Regression Analysis of Niche Lifestyle Keywords and Dataset Inclusion"

# Formulating the model using a string formula
formula = 'in_dataset ~ ' + ' + '.join(keywords_combined.columns.drop('in_dataset'))
```

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```
model = smf.logit(formula=formula, data=keywords_combined)
result = model.fit()

# Extract results for table
df_logistic_regression = pd.DataFrame({
    'coef': result.params,
    'std': result.bse,
    'p_value': result.pvalues,
    'ci_low': result.conf_int()[0],
    'ci_high': result.conf_int()[1]
})

df_to_latex(df_logistic_regression, 'df_logistic_regression',
    ↪ caption=caption)

## Figure df_logistic_fig:
caption = "Logistic Regression Coefficients with Confidence
    ↪ Intervals"

df_logistic_regression['ci'] = list(zip(df_logistic_regression['
    ↪ ci_low'], df_logistic_regression['ci_high']))

df_to_figure(
    df_logistic_regression,
    'df_logistic_fig',
    caption=caption,
    kind='bar',
    y=['coef'],
    y_ci=['ci'],
    y_p_value=['p_value']
)

# SAVE ADDITIONAL RESULTS
additional_results = {
    'Total number of observations': len(df),
    'Number of positive class for in_dataset': df['in_dataset'].
        ↪ sum(),
    'Logistic regression AIC': result.aic
}

with open('additional_results.pkl', 'wb') as f:
    pickle.dump(additional_results, f)
```

C.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
```

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```
        filename: str, caption: str,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    """


def df_to_figure(
        df, filename: str, caption: str,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
    ):
    """
    Save a DataFrame `df` and create a LaTeX figure.
    Parameters, for LaTex embedding of the figure:
    `filename`: Filename for the figure.
    `caption`: Caption for the figure.

    Parameters for df.plot():
    `x`: Column name for x-axis (index by default).
    `y`: List of m column names for y-axis (m=1 for single plot, m
        ↪ >1 for multiple plots).
    `kind`: only bar is allowed.
    `logx` / `logy` (bool): log scale for x/y axis.

    `y_ci`: Confidence intervals for errorbars.
        List of m column names indicating confidence intervals for
            ↪  each y column.
        Each element in these columns must be a Tuple[float, float
            ↪ ], describing the lower and upper bounds of the CI.

     `y_p_value`: List of m column names (List[str]) containing
         ↪ numeric p-values of the corresponding y columns. These
         ↪ numeric values will be automatically converted by
         ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
         ↪ plotted above the error bars.

    If provided, the length of `y_ci`, and `y_p_value` should be
        ↪ the same as of `y`.

    Example:
    Suppose, we have:

    df_lin_reg_longevity = pd.DataFrame({
```

```
    'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
        ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
    'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
        ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
        ↪ [0.0001, ...],
}, index=['var1', ...])

then:
df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
    ↪ caption='Coefficients of ...', kind='bar',
    y=['adjusted_coef', 'unadjusted_coef'],
    y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
    y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
"""
```

C.3 Code Description

C.4 Dataset Preparation and Preprocessing

The dataset, loaded from a CSV file, comprises pairs of textual representations describing speaker personas, from which niche lifestyle keywords are extracted. The textual data provided by ‘Speaker 1‘ and ‘Speaker 2‘ are initially parsed into list structures using Python’s Abstract Syntax Tree (AST) framework to ensure accurate processing of string representations into list formats. This step ensures the integrity and safety of parsing operations while handling string lists.

C.5 Text Vectorization and Keyword Identification

To facilitate the analysis and identification of niche lifestyle keywords, the ‘CountVectorizer‘ tool from the ‘sklearn‘ library is utilized. This tool transforms text data into vectorized format by creating binary indicators for the presence of specific keywords, such as ’vegan’, ’metal’, ’Metallica’, etc. These keywords were selected based on their potential relevance to the hypothesis under investigation. The text data for each speaker is processed separately, after which the keyword occurrences for both speakers are combined to form a single dataset, ensuring any particular keyword is present at most once for any pair of speakers.

C.6 Data Preparation for Logistic Regression

To ready the dataset for logistic regression analysis, a combined DataFrame of keyword occurrences is formed. This DataFrame includes an additional

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'in_dataset' variable, indicating the presence of the speaker pair within the dataset. Keywords with zero variance are removed to ensure computational stability and avoid singular matrix issues in the subsequent regression analysis.

C.7 Logistic Regression Analysis

The core analytical task involves a logistic regression to ascertain the relationship between the presence of niche lifestyle keywords and the inclusion of speaker pairs within the dataset. Utilizing the 'statsmodels' library, the model is specified with the 'in_dataset' variable as the response and the identified keywords as predictors. The regression analysis yields coefficients, standard errors, p-values, and confidence intervals for each keyword predictor, providing insight into which lifestyle choices significantly correlate with dataset inclusion.

C.8 Visualization and Summary of Results

The results of the logistic regression are formatted into a table, which is subsequently exported to a LaTeX format file using a custom utility function 'df_to_latex'. Additionally, a visual representation with confidence intervals of the logistic regression coefficients is produced using another utility function 'df_to_figure', aiding in the interpretability of the analysis. The existence and magnitude of relationships between lifestyle choices and dataset inclusion are clearly illustrated.

C.9 Conclusion and Further Information

Additional results, including the total number of observations, number of positive class occurrences in the 'in_dataset' variable, and the Akaike Information Criterion (AIC) from the logistic regression model, are saved for further scrutiny. These steps are integral in understanding the data and validating the statistical model employed.

C.10 Code Output

df_logistic_fig.pkl

```
              coef       std    p_value     ci_low   ci_high
                                      ci
Intercept   -4.813    0.1851  5.33e-149     -5.175     -4.45
              (-5.175, -4.45)
```

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```
vegan       -6.959     4.945      0.159     -16.65     2.733
             (-16.65, 2.733)
metal       -49.88 2.848e+10          1 -5.582e+10 5.582e+10
    (-5.582e+10, 5.582e+10)
leather     -19.64 9.108e+04          1 -1.785e+05 1.785e+05
    (-1.785e+05, 1.785e+05)
basketball 0.4896    0.9855      0.619     -1.442     2.421
             (-1.442, 2.421)
rock        0.1317    0.9331      0.888     -1.697     1.961
             (-1.697, 1.961)
vegetarian  6.491    0.4108   3.03e-56      5.685     7.296
              (5.685, 7.296)
```

df_logistic_regression.pkl

```
             coef       std    p_value     ci_low   ci_high
Intercept  -4.813    0.1851  5.33e-149     -5.175     -4.45
vegan      -6.959     4.945      0.159     -16.65     2.733
metal      -49.88 2.848e+10          1 -5.582e+10 5.582e+10
leather    -19.64 9.108e+04          1 -1.785e+05 1.785e+05
basketball 0.4896    0.9855      0.619     -1.442     2.421
rock       0.1317    0.9331      0.888     -1.697     1.961
vegetarian  6.491    0.4108   3.03e-56      5.685     7.296
```

additional_results.pkl

```
{
    'Total number of observations': 4000,
    'Number of positive class for in_dataset': 31,
    'Logistic regression AIC': 581.9,
}
```

D LaTeX Table Design

D.1 Code

The LaTeX Table Design was carried out using the following custom code:

```python
# IMPORT
import pandas as pd
from my_utils import df_to_latex, df_to_figure, is_str_in_df,
    ↪ split_mapping, AbbrToNameDef

# PREPARATION FOR ALL TABLES AND FIGURES
all_mapping: AbbrToNameDef = {
    # Rename and provide glossary definitions for any abbreviated
        ↪ or not self-explanatory labels:
```

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```
    'coef': ('Coefficient', 'Estimated coefficient from the
        ↪ logistic regression model'),
    'std': ('Std. Error', 'Standard error of the coefficient'),
    'p_value': ('P-value', 'Probability value associated with the
        ↪ statistical test of the coefficient'),
    'ci_low': ('CI Low', 'Lower bound of the 95% confidence
        ↪ interval for the coefficient'),
    'ci_high': ('CI High', 'Upper bound of the 95% confidence
        ↪ interval for the coefficient'),
    'ci': ('CI', '95% Confidence Interval for the coefficient'),
    'vegan': ('Vegan', 'Indicates mention of veganism'),
    'metal': ('Metal', 'Indicates mention of metal music'),
    'leather': ('Leather', 'Indicates mention of leather'),
    'basketball': ('Basketball', 'Indicates mention of basketball'
        ↪ ),
    'rock': ('Rock', 'Indicates mention of rock music'),
    'vegetarian': ('Vegetarian', 'Indicates mention of
        ↪ vegetarianism'),
}

## Process df_logistic_fig:
df_logistic_fig = pd.read_pickle('df_logistic_fig.pkl')

# Format values:
# Not Applicable

# Rename rows and columns:
# Remove Intercept from the dataframe
df_logistic_fig = df_logistic_fig.drop(index='Intercept')

mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_logistic_fig, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_logistic_fig.rename(columns=abbrs_to_names, index=
    ↪ abbrs_to_names, inplace=True)

# As a figure:
df_to_figure(
    df_logistic_fig, 'df_logistic_fig_formatted',
    caption="Logistic Regression Coefficients for Niche Lifestyle
        ↪ Keywords",
    glossary=glossary,
    kind='bar',
    y=['Coefficient'],
    y_ci=['CI'],
    y_p_value=['P-value'],
    xlabel='Keywords',
    ylabel='Estimated Coefficient'
)
```

Created by data-to-paper (AI)

```
## Process df_logistic_regression:
df_logistic_regression = pd.read_pickle('df_logistic_regression.
    ↪ pkl')

# Format values:
# Not Applicable

# Rename rows and columns:
# Remove Intercept from the dataframe
df_logistic_regression = df_logistic_regression.drop(index='
    ↪ Intercept')

mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_logistic_regression, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_logistic_regression.rename(columns=abbrs_to_names, index=
    ↪ abbrs_to_names, inplace=True)

# As a table:
df_to_latex(
    df_logistic_regression, 'df_logistic_regression_formatted',
    caption="Logistic Regression Analysis of Niche Lifestyle
        ↪ Keywords and Dataset Inclusion",
    glossary=glossary
)
```

D.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
        note: str = None,
        glossary: Dict[Any, str] = None,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    `note`: Note to be added below the table caption.
    `glossary`: Glossary for the table.
    """


def df_to_figure(
        df, filename: str, caption: str,
        note: str = None, glossary: Dict[Any, str] = None,
        x: Optional[str] = None, y: List[str] = None,
```

Created by data-to-paper (AI)

```
    kind: str = 'bar',
    logx: bool = False, logy: bool = False,
    y_ci: Optional[List[str]] = None,
    y_p_value: Optional[List[str]] = None,
    xlabel: str = None, ylabel: str = None,
):
"""
Save a DataFrame `df` and create a LaTeX figure.
Parameters, for LaTex embedding of the figure:
`filename`: Filename for the figure.
`caption`: Caption for the figure.
`note`: Note to be added below the figure caption.
`glossary`: Glossary for the figure.

Parameters for df.plot():
`x`: Column name for x-axis (index by default).
`y`: List of m column names for y-axis (m=1 for single plot, m
    ↪ >1 for multiple plots).
`kind`: only bar is allowed.
`logx` / `logy` (bool): log scale for x/y axis.
`xlabel`: Label for the x-axis.
`ylabel`: Label for the y-axis.

`y_ci`: Confidence intervals for errorbars.
    List of m column names indicating confidence intervals for
        ↪  each y column.
    Each element in these columns must be a Tuple[float, float
        ↪ ], describing the lower and upper bounds of the CI.

 `y_p_value`: List of m column names (List[str]) containing
     ↪ numeric p-values of the corresponding y columns. These
     ↪ numeric values will be automatically converted by
     ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
     ↪ plotted above the error bars.

If provided, the length of `y_ci`, and `y_p_value` should be
    ↪ the same as of `y`.

Example:
Suppose, we have:

df_lin_reg_longevity = pd.DataFrame({
    'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
        ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
    'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
        ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
        ↪ [0.0001, ...],
}, index=['var1', ...])
```

Created by data-to-paper (AI)

```
    then:
    df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
        ↪ caption='Coefficients of ...', kind='bar',
        y=['adjusted_coef', 'unadjusted_coef'],
        y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
        y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
    """


def is_str_in_df(df: pd.DataFrame, s: str):
    return any(s in level for level in getattr(df.index, 'levels',
        ↪  [df.index]) + getattr(df.columns, 'levels', [df.columns
        ↪ ]))

AbbrToNameDef = Dict[Any, Tuple[Optional[str], Optional[str]]]

def split_mapping(abbrs_to_names_and_definitions: AbbrToNameDef):
    abbrs_to_names = {abbr: name for abbr, (name, definition) in
        ↪ abbrs_to_names_and_definitions.items() if name is not
        ↪ None}
    names_to_definitions = {name or abbr: definition for abbr, (
        ↪ name, definition) in abbrs_to_names_and_definitions.
        ↪ items() if definition is not None}
    return abbrs_to_names, names_to_definitions
```

D.3 Code Output

df_logistic_fig_formatted.pkl

```
\begin{figure}[htbp]
\centering
\includegraphics{df_logistic_fig_formatted.png}
\caption{Logistic Regression Coefficients for Niche Lifestyle
    Keywords
Coefficient: Estimated coefficient from the logistic regression
    model.
Std. Error: Standard error of the coefficient.
P-value: Probability value associated with the statistical test of
     the coefficient.
CI Low: Lower bound of the 95\% confidence interval for the
    coefficient.
CI High: Upper bound of the 95\% confidence interval for the
    coefficient.
CI: 95\% Confidence Interval for the coefficient.
Vegan: Indicates mention of veganism.
Metal: Indicates mention of metal music.
Leather: Indicates mention of leather.
Basketball: Indicates mention of basketball.
```

Created by data-to-paper (AI)

```
Rock: Indicates mention of rock music.
Vegetarian: Indicates mention of vegetarianism.
Significance: ns p $>$= 0.01, * p $<$ 0.01, ** p $<$ 0.001, *** p
    $<$ 0.0001.}
\label{figure:df-logistic-fig-formatted}
\end{figure}
% This latex figure presents "df_logistic_fig_formatted.png",
% which was created from the df:
%
% index,"Coefficient","Std. Error","P-value","CI Low","CI High","
    CI"
% "Vegan",-6.959,4.945,0.159,-16.65,2.733,(-16.65, 2.733)
% "Metal",-49.88,2.848e+10,1,-5.582e+10,5.582e+10,(-5.582e+10,
    5.582e+10)
% "Leather",-19.64,9.108e+04,1,-1.785e+05,1.785e+05,(-1.785e+05,
    1.785e+05)
% "Basketball",0.4896,0.9855,0.619,-1.442,2.421,(-1.442, 2.421)
% "Rock",0.1317,0.9331,0.888,-1.697,1.961,(-1.697, 1.961)
% "Vegetarian",6.491,0.4108,<1e-06,5.685,7.296,(5.685, 7.296)
%
% To create the figure, this df was plotted with the command:
%
% df.plot(kind='bar', y=['Coefficient'], xlabel='Keywords', ylabel
    ='Estimated Coefficient')
%
% Confidence intervals for y-values were then plotted based on
    column: ['CI'].
%
% P-values for y-values were taken from column: ['P-value'].
%
% These p-values were presented above the data points as stars (
    with significance threshold values indicated in the figure
    caption).
```

df_logistic_regression_formatted.pkl

```
\begin{table}[h]
\caption{Logistic Regression Analysis of Niche Lifestyle Keywords
    and Dataset Inclusion}
\label{table:df-logistic-regression-formatted}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{%
\begin{tabular}{lrrlrr}
\toprule
 & Coefficient & Std. Error & P-value & CI Low & CI High \\
\midrule
\textbf{Vegan} & -6.959 & 4.945 & 0.159 & -16.65 & 2.733 \\
```

Created by data-to-paper (AI)

```
\textbf{Metal} & -49.88 & 2.848e+10 & 1 & -5.582e+10 & 5.582e+10
    \\
\textbf{Leather} & -19.64 & 9.108e+04 & 1 & -1.785e+05 & 1.785e+05
     \\
\textbf{Basketball} & 0.4896 & 0.9855 & 0.619 & -1.442 & 2.421 \\
\textbf{Rock} & 0.1317 & 0.9331 & 0.888 & -1.697 & 1.961 \\
\textbf{Vegetarian} & 6.491 & 0.4108 & $<$1e-06 & 5.685 & 7.296 \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item \textbf{Coefficient}: Estimated coefficient from the
    logistic regression model
\item \textbf{Std. Error}: Standard error of the coefficient
\item \textbf{P-value}: Probability value associated with the
    statistical test of the coefficient
\item \textbf{CI Low}: Lower bound of the 95\% confidence interval
     for the coefficient
\item \textbf{CI High}: Upper bound of the 95\% confidence
    interval for the coefficient
\item \textbf{Vegan}: Indicates mention of veganism
\item \textbf{Metal}: Indicates mention of metal music
\item \textbf{Leather}: Indicates mention of leather
\item \textbf{Basketball}: Indicates mention of basketball
\item \textbf{Rock}: Indicates mention of rock music
\item \textbf{Vegetarian}: Indicates mention of vegetarianism
\end{tablenotes}
\end{threeparttable}
\end{table}
```